

Ferric derisomaltose versus ferric carboxymaltose: a systematic review and meta-analysis of head-to-head randomised controlled trials on hypophosphataemia, osteomalacia, hypersensitivity and cardiovascular events

Filippo Buscemi^{1,*} Primo Buscemi²

¹ UOC Medicina Trasfusionale, Presidio Ospedaliero “San Giovanni di Dio”, ASP di Agrigento, Italy

² SOC Coordinamento Sanitario Servizi Zona Sud-Est, Azienda USL Toscana Centro, Firenze, Italy

* Corresponding author: filippo.buscemi@aspag.it

Abstract

Background. Ferric derisomaltose (FDI) and ferric carboxymaltose (FCM) are high-dose intravenous iron formulations thought to differ mainly in the risk of hypophosphataemia. Previous syntheses relied mainly on network, indirect or study-level comparisons. We synthesised only randomised trials that compared the two drugs directly.

Methods. We searched PubMed, Europe PMC, Crossref, ClinicalTrials.gov, the EU Clinical Trials Register and CTIS without date limits, and did citation searching. Embase, Cochrane CENTRAL and WHO ICTRP were not searched. We included randomised controlled trials in adults with iron deficiency that randomised participants directly between an FDI arm and an FCM arm. Title and abstract screening, extraction, risk-of-bias and certainty assessments were done in duplicate by two independent large-language-model agents. An author decided every discordance and every eligibility decision, on proposals prepared by a single agent, and verified every data row selected for analysis; concordant exclusions at title and abstract, and concordant risk-of-bias and GRADE judgements, were not reviewed. The primary outcome was hypophosphataemia (serum phosphate <0.65 mmol/L) within the primary window (day 35 ± 7). We used random-effects (REML with Hartung–Knapp–Sidik–Jonkman), Mantel–Haenszel or Peto models, RoB 2 and GRADE.

Results. We included 10 trials (36 reports; 1940 participants randomised to FDI and 1945 to FCM). FDI may reduce the incidence of hypophosphataemia compared with FCM (4 trials, 364 participants; risk ratio 0.12, 95% CI 0.085 to 0.17; 680 per 1000 with FCM versus 81 with FDI; low-certainty evidence). No participant receiving FDI had severe hypophosphataemia in the trials reporting it (Peto odds ratio 0.11, 0.037 to 0.34; low certainty), and FDI may reduce hypophosphataemia persisting at the end of the primary window (risk ratio 0.055, 0.018 to 0.17; low certainty). Evidence on serious or severe hypersensitivity (risk ratio 0.32, 0.033 to 3.02), any hypersensitivity or infusion reaction, and serious cardiovascular adverse events was very uncertain. No trial reported osteomalacia, fractures or adjudicated cardiovascular events. Of 30 outcome-level risk-of-bias judgements, 27 had some concerns and 3 were at high risk. All trials contributing to the pooled estimates for hypophosphataemia, persistent hypophosphataemia and change in phosphate were funded by the manufacturer of FDI.

Conclusions. In head-to-head randomised trials, FDI may reduce the incidence of hypophosphataemia compared with FCM. Evidence on hypersensitivity and cardiovascular safety is very uncertain, and no head-to-head trial has reported osteomalacia or fractures.

Registration. The protocol was frozen in the project’s version-controlled repository (commit fc19a3a) before the systematic searches and arm-level data extraction, and is publicly available; it was not registered in a review registry.

1 Introduction

Ferric derisomaltose (FDI, formerly iron isomaltoside 1000) and ferric carboxymaltose (FCM) allow the replacement of large iron deficits in one or two infusions. Their comparison has centred on hypophosphataemia, which the phosphate-focused trials attributed to a rise in intact fibroblast growth factor 23 after FCM [1]. Other safety questions concern hypersensitivity reactions and cardiovascular events.

Several syntheses have addressed these questions. Most pooled evidence through network meta-analysis or indirect comparison [2–5], or through study-level comparisons [6, 7]. Indirect estimates rest on the assumption that trials against different comparators are exchangeable. That assumption is hard to defend here, because the populations, doses and definitions of hypophosphataemia vary widely between studies.

We therefore restricted this review to randomised controlled trials that compared FDI and FCM directly. The outcomes were hypophosphataemia, osteomalacia and fractures, hypersensitivity, and cardiovascular events. The review was run in a version-controlled evidence repository in which every aggregate number is produced by a script from verified arm-level rows. The methods of that repository will be described in a separate report (in preparation).

2 Methods

2.1 Protocol, registration and reporting

The protocol was written in the project repository and frozen at commit `fc19a3a` (29 September 2026). It was frozen before the systematic searches, before arm-level data extraction and before any aggregate calculation; before freezing, a dry run of the PubMed query had been done to check that it carried no date limit. The repository, however, already contained unstructured notes and extractions that were not organised by arm for some candidate trials, and part of the PubMed records were already in it; the choice of the primary threshold was motivated with those data. The protocol was not registered in a review registry such as PROSPERO or OSF Registries; the frozen version, the current version with its deviation log and the commit history of the review files are in the public data repository (Declarations). All deviations from the frozen protocol are listed in Table 1. Reporting follows PRISMA 2020 [8].

2.2 Eligibility criteria

We included randomised controlled trials that randomised adults (≥ 18 years) with iron deficiency, with or without anaemia and of any cause, directly between an FDI arm and an FCM arm, at any dose and schedule. In multi-arm trials, only the FDI and FCM arms were used. Mixed adult–paediatric trials were eligible if adult data were separable or if minors were fewer than 10% of those randomised.

We excluded:

- non-randomised and quasi-randomised studies;
- indirect comparisons;
- open-label extensions without randomisation;
- trials that co-administered another intravenous iron in the same arm.

Crossover trials were eligible using the first period only; a crossover trial without separately reported first-period data would be listed and not pooled. There were no limits on date, language or publication status: conference abstracts and results posted in trial registers were eligible if they reported arm-level data.

2.3 Outcomes

The primary outcome was the incidence of hypophosphataemia, defined as serum phosphate <0.65 mmol/L (<2.0 mg/dL) at any measurement within the primary window. The primary window ran from the first dose to day 35 (± 7), or to the trial's primary time point if it fell between days 28 and 42.

Secondary outcomes were:

- severe hypophosphataemia (≤ 0.32 mmol/L);
- hypophosphataemia at a wide threshold (<0.80 mmol/L, a threshold of 0.81 mmol/L being grouped here);
- persistent hypophosphataemia (<0.65 mmol/L at the last visit of the primary window);
- change in serum phosphate from baseline to day 35;
- osteomalacia or fragility fractures;
- serious or severe hypersensitivity;
- any hypersensitivity or infusion reaction;
- cardiovascular events: adjudicated events as the main outcome, and serious adverse events in the

MedDRA *Cardiac disorders* and *Vascular disorders* system organ classes as a sensitivity outcome. A trial whose phosphate threshold differed from the protocol thresholds entered the outcome whose protocol threshold was the nearest one not above the trial's threshold (for example, 0.83 mmol/L entered the outcome at 0.80 mmol/L); a trial whose threshold was below every protocol threshold of an outcome did not enter it. The difference was tested in sensitivity analyses. Events over the longest available follow-up were analysed separately as an extended window.

2.4 Information sources and search

We searched the following sources on 29 September 2026 with no date limit:

- PubMed/MEDLINE, with a query combining all synonyms of both drugs (including the former name iron isomaltoside) and no design filter;
 - Europe PMC, for preprints and records not in MEDLINE;
 - Crossref, for conference abstracts published in journal supplements;
 - ClinicalTrials.gov (API v2), the EU Clinical Trials Register and the EU CTIS.
- Embase, Cochrane CENTRAL and WHO ICTRP, planned in the protocol, were not searched; the search strings prepared for them are in the repository. As other methods we used:
- a supplementary search in the Consensus engine;
 - automated citation searching through the Europe PMC API: the references of the reports included at the time of the search and of related reviews, and the citations of those included reports; one conference abstract was added by hand from the reference list of an included report.

The citation search used a widened filter: both drugs, or one drug or intravenous iron in a randomised record. We adopted this filter after finding that the published protocol of one included trial did not name either drug in its title or abstract. Full search strings and dates are in the repository.

2.5 Selection process

Records were de-duplicated by PMID, DOI and registry number, and one duplicate was merged by hand on the title. Titles and abstracts were screened in duplicate by two independent large-language-model agents (Claude, Anthropic) applying written operational criteria, blind to each other’s decisions. Records that both agents excluded were excluded without author review; every discordant or uncertain record was decided by an author. Agreement was assessed with Cohen’s kappa [9] across three categories (include, exclude, uncertain). Full texts, registry records and conference abstracts were then assessed for eligibility; every eligibility decision was taken by an author, who read structured evidence sheets prepared by a single agent run and approved or amended its proposals, in blocks. Reports were grouped into trials by registry number or, for unregistered trials, by an identifier based on the primary report.

2.6 Data collection

Two independent agents extracted arm-level data for each trial with a written codebook. Every source of the trial was used: articles, supplements, registry results and abstracts. A script compared the two extractions cell by cell, and an author resolved every discordance. An author then checked against the source text every row selected for analysis (68 of 159 extracted rows). The preferred analysis population was randomised participants who received at least one dose. We never derived event counts from percentages. Zero event counts were taken only when the source stated them, including from complete lists of serious adverse events in registries and from explicit statements that no serious adverse events occurred.

2.7 Risk of bias

We assessed risk of bias with RoB 2 [10] for each outcome of each trial, for the effect of assignment. Two independent agents answered the signalling questions, quoting the supporting text and its location. A script checked that each domain judgement followed the RoB 2 algorithm from the answers, allowing departures only with a written justification, and an author resolved discordances. The overall judgement was the worst domain judgement, or high when four or more domains had some concerns and the report was internally inconsistent; this rule was added to the assessment guide while resolving a discordance on the trial to which it was then applied (Table 1). Concordant judgements were accepted without author review.

2.8 Synthesis methods

All aggregate numbers were produced by a single R script (R 4.5.0, metafor 5.2-1 [11], lme4 2.0-6 [12]). The script reads only the verified arm-level rows, together with the trial characteristics and risk-of-bias judgements used in sensitivity analyses and the subgroup classification. The run reported here corresponds to commit d75c923 of the working repository and input checksum a0346f9973f5; the public data repository reproduces the same outputs. Effects are expressed for FDI relative to FCM.

Dichotomous outcomes. The main model for phosphate outcomes was a random-effects model on the log risk ratio, with REML and the Hartung–Knapp–Sidik–Jonkman correction [13], without continuity correction. With fewer than five trials, a Mantel–Haenszel fixed-effect estimate [14] is

shown alongside. Double-zero trials were excluded from relative measures and kept in risk differences. Otherwise, zero cells were handled as follows:

- when a trial with events had an arm with no events (or with all participants having the event), which would remove it from the REML model, the Mantel–Haenszel model became the main model;
- when one arm had no events in any trial, the risk ratio is undefined, and the main measures were the Peto odds ratio and the Mantel–Haenszel risk difference;
- when fewer than two trials had events, only the Mantel–Haenszel risk difference, which keeps double-zero trials, was pooled (a rule not in the protocol); when no trial had events, no effect was estimated.

For rare outcomes (hypersensitivity and cardiovascular events), the main model was the Mantel–Haenszel risk ratio, with the risk difference alongside. The Peto odds ratio and, with at least three trials with events (the protocol required three trials), a generalised linear mixed model were sensitivity analyses [15].

Pooling rules. An outcome was pooled when at least two trials reported it. For osteomalacia and serious or severe hypersensitivity, at least two trials also had to report at least one event.

Continuous outcome. The change in phosphate was pooled as a mean difference in mmol/L. Values at day 35 would have been pooled with changes from baseline, with a sensitivity analysis restricted to changes; only changes were available for pooling.

Heterogeneity and absolute effects. We report τ^2 , I^2 and Q , and a prediction interval with at least three trials. Absolute effects were computed from the crude pooled risk in the FCM arms of the trials in each model.

Sensitivity analyses. The prespecified sensitivity analyses were:

- trials at low overall risk of bias only;
- fixed-effect versus random-effects models;
- trials with the exact protocol threshold only;
- exclusion of trials with manufacturer funding or declared manufacturer involvement;
- exclusion of data available only from registers or abstracts;
- a continuity correction of 0.5.

The planned sensitivity analysis on the phosphate nadir was not performed, because nadir values were not extracted.

Subgroups. Exploratory subgroups by population and by dose schedule were analysed. For dose schedule we compared single-dose FDI against split-dose FCM with the same schedule in both arms; the protocol had specified equivalent cumulative doses, which the extracted data did not allow. Subgroups were analysed with the main model when at least two subgroups had at least two trials each, with a test of difference between subgroup estimates. The classification of trials is in the repository.

Publication bias. Funnel plots were not used, because no outcome had ten or more trials. Instead, we compared the registered head-to-head trials identified in ClinicalTrials.gov and the EU Clinical Trials Register with their publications; the protocol had planned this for trials registered as completed, including ICTRP, which was not searched.

2.9 Certainty of evidence

We rated certainty with GRADE [16], using a minimally contextualised approach with the null effect as threshold. Imprecision took the optimal information size into account [17]. Outcomes without a single pooled estimate were rated narratively [18]. Two independent agents rated each outcome; the agent resolved the four discordances, and an author confirmed them after the manuscript was drafted (Table 1); concordant ratings were accepted without author review. Summary statements follow the GRADE wording for informative statements [19].

2.10 Use of large language models

Agents made the first pass of every step. Title and abstract screening, extraction, risk of bias and GRADE used two independent runs; the eligibility evidence sheets were prepared by a single run. Concordant exclusions at title and abstract screening, and concordant risk-of-bias and GRADE judgements, were final without author review. All other final decisions were taken by an author: discordant and uncertain screening records, eligibility (approving or amending the agent’s proposals), every data row selected for analysis, and discordant risk-of-bias and GRADE judgements (the GRADE ones after the manuscript was drafted). No agent produced an aggregate number. Agent outputs, or summaries of them, are archived in the repository together with the scripts that compared them.

3 Results

3.1 Study selection

Databases and registers yielded 295 records (Figure 1). After removal of 47 duplicates and 1 record not importable, 247 records were screened and 189 excluded.

Of 63 reports sought for retrieval, 5 were abstracts identified inside conference-abstract volumes; 7 could not be retrieved, 21 were excluded after full assessment, and 1 registered trial without results (NCT06350955) is awaiting classification.

Citation searching identified 1124 records, 951 of which were new. Of the 190 that passed the widened filter (including one abstract added by hand from a reference list), 2 were included. The supplementary Consensus search found 4 new records, all also retrieved by Crossref.

Across both branches, screening agreement between the two agents was 98.4% (kappa 0.93, 437 paired decisions). In total, 36 reports of 10 trials were included.

3.2 Characteristics of the included trials

Table 2 describes the 10 trials. Of these, 3 were open-label, 3 were described as double-blind and 1 as partially masked; 5 were funded wholly or partly by Pharmacosmos, the manufacturer of FDI; 1 involved Vifor, the manufacturer of FCM, without a funding declaration; and 4 had no manufacturer funding (in one of these, commercially available study drugs were purchased from a distributor). A phase I pharmacokinetic trial [20] reported none of the review outcomes.

Table 1: Deviations from the frozen protocol (commit `fc19a3a`).

Topic	Deviation	Decided
Screening	Title/abstract screening in duplicate by two agents instead of single; concordant exclusions final	Before screening
Screening criteria	Written operational criteria and a dedicated screening agent; criteria revised twice (after the PubMed screening; rule for conference volumes at eligibility)	Before screening; revisions during selection
Eligibility	Evidence sheets prepared by a single agent run; decisions approved by an author in blocks	During selection
Awaiting classification	Registered trials without results kept as “awaiting classification”	Before screening
Sources not searched	Embase, CENTRAL and WHO ICTRP not searched; publication-bias rating left as provisional in GRADE	–
Sources	Crossref added as a systematic source for conference abstracts (relevance ranking not guaranteed stable over time)	After screening the PubMed records, before screening the other sources
Citation search	Automated through the Europe PMC API instead of manual; widened filter (one drug or intravenous iron in a randomised record)	After one included protocol was missed by the main filter
Extraction	Written codebook, a second file for trial characteristics and service columns	Before extraction
Change in phosphate	Main time point fixed at day 35, with the nadir as a sensitivity analysis; the nadir analysis was then not performed (nadir not extracted)	Day 35 before any calculation; nadir recorded as not performed after the analysis
Zero cells	Mantel–Haenszel as main model when zero cells remove trials from REML; Peto odds ratio and risk difference when one arm has no events	After seeing which models were estimable
Pooling rule	Script corrected to the protocol (it required two trials with events for every outcome); risk difference as the main measure when fewer than two trials have events, a new rule; no estimate when no trial has events	After seeing the results
Threshold rule	ExplorIRON-CKD removed from severe hypophosphataemia, its threshold being below every protocol threshold of that outcome (application of the protocol)	After seeing the results
Any hypersensitivity	Not pooled, because of different definitions and effects in opposite directions	After seeing the results
Subgroups	Omitted in the first run; dose subgroup defined as single-dose FDI against split-dose FCM versus the same schedule (protocol: equivalent cumulative doses); classification file read by the script, written after the data were known	After seeing the results
Publication bias	Registered versus published trials compared descriptively (no funnel plot), from the registries in the database and not restricted to trials registered as completed	After seeing the results
Assessment guides	Written guides for RoB 2 and GRADE (null-threshold imprecision, optimal information size, provisional publication-bias rating); duplicate GRADE ratings; RoB 2 guide and extraction codebook revised twice; the rule for a high overall judgement by accumulation, and the codebook rule on zeros from complete adverse-event lists, were written on trials to which they were then applied	Before each assessment; revisions during the assessment
Software	Package versions recorded at every run instead of a lock file	Before analysis, after extraction
GLMM	Fitted with at least three trials with events (protocol: at least three trials)	Before analysis, after extraction (event counts known)
Primary outcome, extended window	A double-zero trial, whose whole-study zero was already used for the primary window, also used for the extended window	After seeing the results
GRADE adjudication	The four discordances between the two GRADE runs were first resolved by the agent in the consensus; an author	During the assessment; author

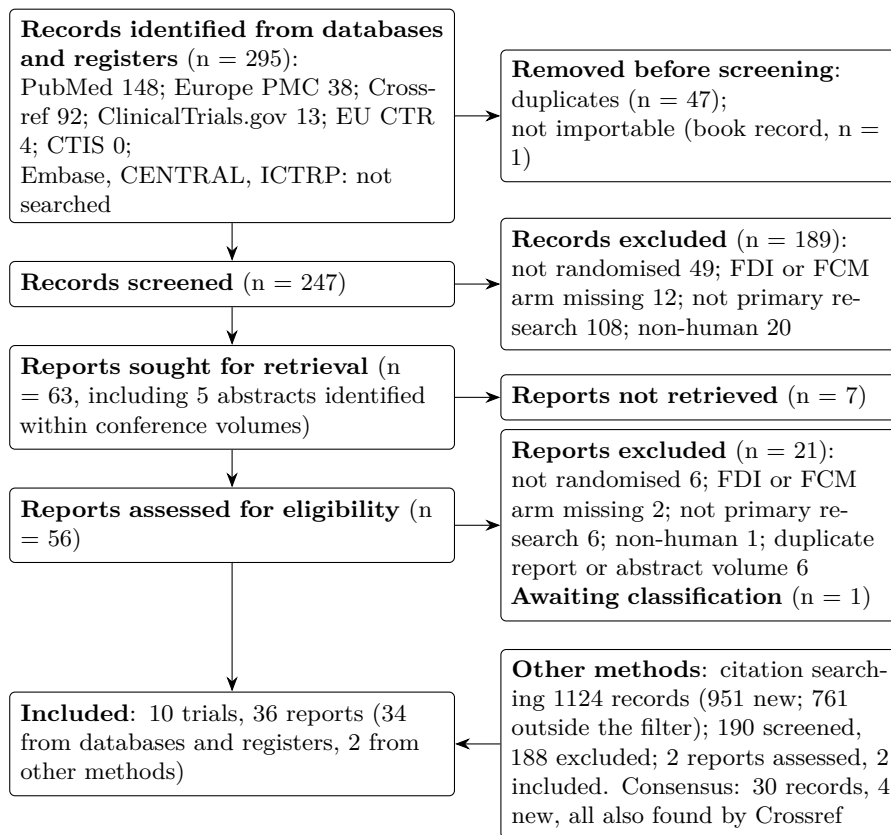


Figure 1: PRISMA 2020 flow diagram. All counts are generated from the project database and search log.

Table 2: Characteristics of the included trials (only the FDI and FCM arms are shown). n = randomised. Follow-up in days; NR = not reported.

Trial	Country	Population	Blinding	n FDI / FCM	FDI dose	FCM dose	Follow-up	Funding
PHOSPHARE-IDA04 [1]	USA	Adults with iron-deficiency anaemia of various causes, intolerant of or unresponsive to oral iron; mostly women with gynaecological blood loss	Open-label	62 / 61	1000 mg single infusion, day 0	750 mg on days 0 and 7 (1500 mg)	35	Industry (Pharmacosmos)
PHOSPHARE-IDA05 [1]	USA	Adults with iron-deficiency anaemia of various causes, intolerant of or unresponsive to oral iron	Open-label	61 / 61	1000 mg single infusion, day 0	750 mg on days 0 and 7 (1500 mg)	35	Industry (Pharmacosmos)
PHOSPHARE-IBD [21]	Austria, Denmark, Germany, Sweden, UK	Adults with inflammatory bowel disease and iron-deficiency anaemia	Double-blind	49 / 48	1000 mg day 0 + 500 or 1000 mg day 35 (1500 or 2000 mg)	1000 mg day 0 + 500 or 1000 mg day 35 (1500 or 2000 mg)	70	Industry (Pharmacosmos)
HOME aFers [22]	Germany	Adult women with iron-deficiency anaemia from uterine bleeding	Double-blind	13 / 13	20 mg/kg, max 1000 mg, single infusion	20 mg/kg, max 1000 mg, single infusion	35	Industry (Pharmacosmos)
ExplorIRON-CKD [23]	UK	Adults with non-dialysis CKD stage 3a–5 and iron deficiency	Double-blind	14 / 12	1000 mg day 0 + 500 or 1000 mg at 1 month (1500 or 2000 mg)	1000 mg day 0 + 500 or 1000 mg at 1 month (1500 or 2000 mg)	90	Mixed (Pharmacosmos)
Bombač Tavčar 2024 [24]	Slovenia	Women with post-partum anaemia within 48 h of delivery (three arms, including oral iron)	Open-label	100 / 100	Modified Ganzoni dose (1000–2000 mg), max 1500 mg/day	Modified Ganzoni dose (1000–2000 mg), max 1000 mg/day	42	None
RAPIDIIRON [25]	India	Pregnant women at 14–17 weeks with moderate iron-deficiency anaemia (three arms, including oral iron)	Partially masked (IV arms; pharmacist unblinded)	1456 / 1462	20 mg/kg, max 1000 mg, single infusion	20 mg/kg, max 1000 mg, single infusion	NR	Foundation (CIFF)
Yanamandra 2026 [26]	India	Adults 18–80 years with iron-deficiency anaemia and oral-iron intolerance	Single-blind (patients)	148 / 151	1000 mg iron isomaltoside (i3R), single infusion	1000 mg, single infusion	168	None
Murtineedi 2025 [27]	India	Women 18–50 years with iron-deficiency anaemia (three arms, including iron sucrose)	Not reported	29 / 29	Ganzoni dose, max 1500 mg per infusion	Ganzoni dose, max 1000 mg per session; further doses on days 7 and 14 if needed	28	None
Garbowski 2020 [20]	Not reported	Non-anaemic hypoferraemic healthy adults (phase I PK/PD, four arms)	Single-blind	8 / 8	200 mg single infusion	200 mg single infusion	14	Not declared (Vifor involvement)

3.3 Risk of bias

Table 3 shows the 30 outcome-level judgements: 0 at low risk, 27 with some concerns and 3 at high risk. The commonest concern in the trials contributing to the phosphate outcomes was that safety analyses followed the drug received rather than the assignment, for a few patients. Of the three judgements at high risk, two (one trial) were high because of the randomisation domain, and one was rated high because four domains had some concerns in an internally inconsistent report.

Table 3: Risk of bias (RoB 2) by trial and outcome. D1 randomisation; D2 deviations from intended interventions; D3 missing outcome data; D4 measurement of the outcome; D5 selection of the reported result. ● low; ▲ some concerns; ■ high.

Trial	Outcome	D1	D2	D3	D4	D5	Overall
PHOSPHARE-IDA04	P1	●	▲	●	●	●	▲
PHOSPHARE-IDA04	S1	●	▲	●	●	▲	▲
PHOSPHARE-IDA04	S3	●	▲	●	●	●	▲
PHOSPHARE-IDA04	S4	●	▲	●	●	●	▲
PHOSPHARE-IDA04	S6	●	▲	●	▲	●	▲
PHOSPHARE-IDA04	S8b	●	▲	●	▲	●	▲
PHOSPHARE-IDA05	P1	●	▲	●	●	●	▲
PHOSPHARE-IDA05	S1	●	▲	●	●	▲	▲
PHOSPHARE-IDA05	S3	●	▲	●	●	●	▲
PHOSPHARE-IDA05	S4	●	▲	●	●	●	▲
PHOSPHARE-IDA05	S6	●	▲	●	▲	●	▲
PHOSPHARE-IDA05	S8b	●	▲	●	▲	●	▲
PHOSPHARE-IBD	P1	●	▲	●	●	●	▲
PHOSPHARE-IBD	S3	●	▲	▲	●	●	▲
PHOSPHARE-IBD	S6	●	▲	●	●	▲	▲
PHOSPHARE-IBD	S8b	●	▲	▲	●	●	▲
HOMe aFers	P1	●	●	●	●	▲	▲
HOMe aFers	S3	●	●	●	●	▲	▲
HOMe aFers	S4	●	●	●	●	▲	▲
ExplorIRON-CKD	P1	●	●	▲	●	●	▲
ExplorIRON-CKD	S2	●	●	▲	●	▲	▲
Bombač Tavčar 2024	S1	●	▲	▲	●	▲	▲
Bombač Tavčar 2024	S2	●	▲	▲	●	▲	▲
Bombač Tavčar 2024	S6	●	●	●	▲	▲	▲
Bombač Tavčar 2024	S7	●	●	●	▲	▲	▲
RAPIDIRON	S6	●	▲	●	●	●	▲
RAPIDIRON	S7	●	▲	●	●	▲	▲
Yanamandra 2026	S6	■	●	▲	▲	▲	■
Yanamandra 2026	S8b	■	●	▲	▲	▲	■
Murtineedi 2025	S6	▲	▲	●	▲	▲	■

3.4 Hypophosphataemia

In the primary window, FDI reduced the incidence of hypophosphataemia (4 trials, 364 participants; risk ratio 0.12, 95% CI 0.085 to 0.17; $I^2 = 0\%$; Figure 2). A fifth trial had no events in either arm. The risk was 680 per 1000 with FCM and 81 per 1000 with FDI, a difference of -599 per 1000 (-622 to -566). The estimate was unchanged by the Mantel–Haenszel model (0.12) and by the fixed-effect model. The sensitivity analyses excluding manufacturer-funded trials and restricted to low risk of bias could not be performed, because no trial qualified. In the exploratory subgroups by dose schedule, the risk ratio was 0.11 (0.087 to 0.13) with single-dose FDI against split-dose FCM

and 0.15 (0.013 to 1.63) with the same schedule in both arms (two trials each; test for difference $p = 0.09$). Subgroups by population could not be formed, because only one population had two or more trials. RAPIDIRON reported hypophosphataemia at an early visit (25/1422 with FDI versus 62/1432 with FCM) and at 6 weeks after delivery (21/1341 versus 33/1325), but did not state the threshold in the article or the published protocol, so these data could not be assigned to an outcome and were not used. In the extended window two trials reported the outcome, one with events (6/48 versus 29/49) and one without (0/14 versus 0/12); the risk difference was -0.37 (-0.51 to -0.23).

No participant receiving FDI had severe hypophosphataemia (0/217 versus 13/210). The three trials used different thresholds: 0.32 mmol/L in the two PHOSPHARE-IDA trials and 0.34 mmol/L in the post-partum trial, which measured phosphate only at 6 weeks. ExplorIRON-CKD, whose threshold was below 0.32 mmol/L and which had no events, did not enter this outcome under the threshold rule. The Peto odds ratio, from the two trials with events, was 0.11 (0.037 to 0.34). The risk difference was -0.063 (-0.096 to -0.03 ; 3 trials) and -0.11 (-0.17 to -0.055) when restricted to the exact threshold. In both trials with events, this outcome was a post hoc analysis in the registry.

Hypophosphataemia persisting at the last visit of the primary window was less frequent with FDI (Mantel–Haenszel risk ratio 0.055, 0.018 to 0.17; 4 trials; $I^2 = 56\%$). The REML model alongside lost one trial to zero cells and was imprecise (0.17, 0.0056 to 5.34). In exploratory subgroups, the risk ratio was 0.02 (0.0028 to 0.14) with single-dose FDI against split-dose FCM and 0.40 (0.084 to 1.91) with the same schedule (test for difference $p = 0.019$); with two trials per subgroup and a classification written after the data were known, this is hypothesis-generating only.

At day 35, serum phosphate changed little from baseline with FDI and fell with FCM (mean difference 0.36 mmol/L, 0.24 to 0.49; 2 trials, both funded by the manufacturer of FDI). Four further trials reported phosphate in forms that could not be pooled: values at day 35 without the number of participants per visit, medians, and a least-squares mean change without dispersion in an abstract. In the trial without manufacturer funding, which measured phosphate only at 6 weeks, median values were almost identical in the two arms.

Hypophosphataemia at the wide threshold was reported by single trials only: 2/93 versus 3/95 in the post-partum trial (threshold 0.83 mmol/L, at 6 weeks) and, in the extended window, 1/14 versus 1/12 in ExplorIRON-CKD (threshold 0.81 mmol/L, episodes reported as adverse events).

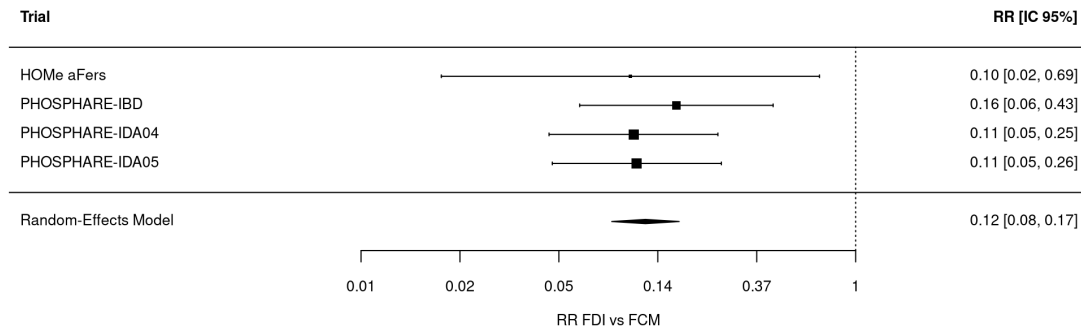
3.5 Hypersensitivity

Serious or severe hypersensitivity reactions were rare in the primary window (1/1708 with FDI versus 3/1697 with FCM). The Mantel–Haenszel risk ratio was 0.32 (0.033 to 3.02; 3 trials with events), and the Mantel–Haenszel risk difference, including double-zero trials, was -0.0012 (-0.0036 to 0.0011; 5 trials). Definitions ranged from protocol-defined serious or severe reactions, whose criteria were not available to us, to externally adjudicated infusion-related serious adverse events without severity criteria.

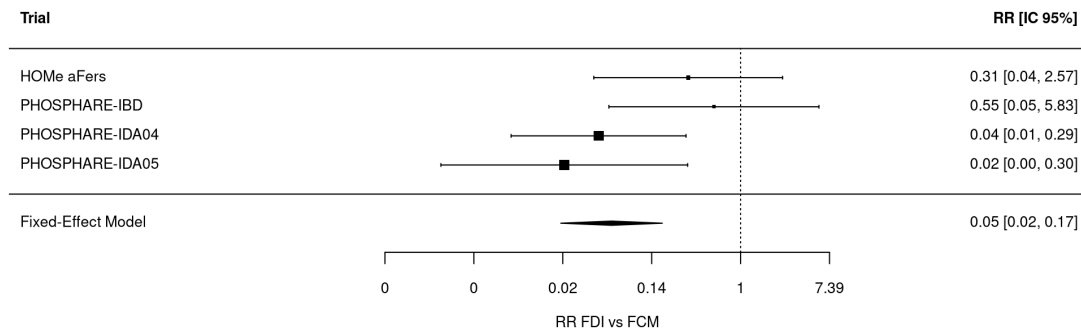
Any hypersensitivity or infusion reaction was not pooled. The two trials used different definitions and pointed in opposite directions:

- 1/100 versus 4/100 in the post-partum trial;
- 340/1454 versus 70/1451 in RAPIDIRON (risk ratio 4.85, 3.79 to 6.20).

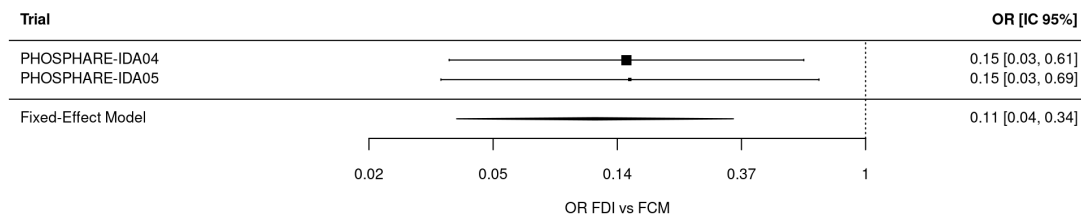
P1, finestra primaria — REML + HKSJ, senza correzione



S3, finestra primaria — MH effetti fissi, senza correzione



S1, finestra primaria — Peto



S4, finestra primaria — MD in mmol/L, REML + HKSJ

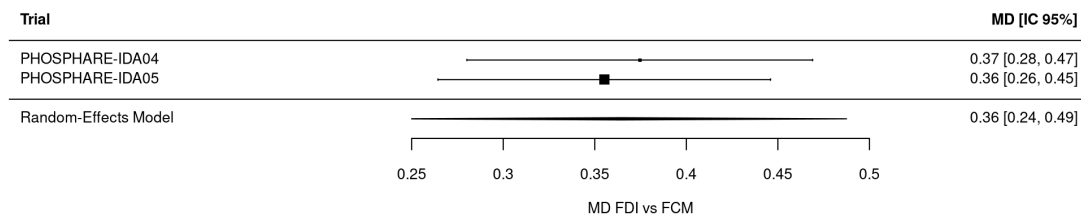


Figure 2: Forest plots of the main analyses: hypophosphataemia (REML–HKSJ, risk ratio), persistent hypophosphataemia (Mantel–Haenszel, risk ratio), severe hypophosphataemia (Peto odds ratio) and change in serum phosphate at day 35 (mean difference, mmol/L). Effects are FDI relative to FCM.

3.6 Cardiovascular events, osteomalacia and fractures

No trial reported adjudicated cardiovascular events. Serious cardiac or vascular adverse events were zero in both arms of the two trials reporting them in the primary window (0/125 versus 0/117), so no effect could be estimated. In the extended window the risk difference was 0.00011 (-0.014 to 0.015 ; 2 trials, 1/188 versus 1/189). ExplorIRON-CKD listed one serious atrial fibrillation event in the FDI arm by event rather than by patient, which was not extracted, and one trial prespecified an adjudicated cardiovascular composite that it did not report. No trial reported osteomalacia or fractures.

3.7 Registered and published trials

Of the 7 included trials whose registry records we retrieved, all 7 had been published. One registered head-to-head trial (NCT06350955) had no results; its registry status was unknown, last reported as recruiting, and its estimated completion date had passed only recently. It is listed as awaiting classification. The registry of one further included trial (CTRI) was not searched, and ICTRP was not searched.

3.8 Certainty of the evidence

Table 4 summarises the findings. Hypophosphataemia (in both windows), severe and persistent hypophosphataemia and the change in phosphate were of low certainty, downgraded for risk of bias and imprecision. For the primary outcome, imprecision was downgraded only because the number of events was below the optimal information size, while the confidence interval was far from the null. All other outcomes, including hypophosphataemia at the wide threshold, were of very low certainty.

Table 4: Summary of findings: FDI compared with FCM in adults with iron deficiency. Risks per 1000 participants. The FCM risk is the crude pooled risk in the FCM arms of the trials in each model; the FDI risk is model-derived from it and from the relative effect (for severe hypophosphataemia no event was observed with FDI).

Outcome	Trials (participants)	Risk with FCM	Risk with FDI	Difference per 1000 (95% CI)	Relative effect (95% CI)	Certainty
Hypophosphataemia (<0.65 mmol/L), primary window	4 (364)	680	81	−599 (−622 to −566)	RR 0.12 (0.085 to 0.17)	Low
Hypophosphataemia (<0.65 mmol/L), extended window	2 (123)	475		−369 (−510 to −227)	RD −0.37 (−0.51 to −0.23)	Low
Severe hypophosphataemia (<0.32 mmol/L), primary window	2 (239) (OR); 3 (427) (RD)	113 (OR); 62 (RD)	14	−99 (−108 to −71); −63 (−96 to −30)	OR 0.11 (0.037 to 0.34); RD −0.063 (−0.096 to −0.03)	Low
Hypophosphataemia, wide threshold (<0.80 mmol/L), primary window	1 (188)				Not pooled: Bombač Tavčar 2024 2/93 vs 3/95	Very low
Hypophosphataemia, wide threshold (<0.80 mmol/L), extended window	1 (26)				Not pooled: ExplorIRON-CKD 1/14 vs 1/12	Very low
Persistent hypophosphataemia, primary window	4 (342)	316	17	−298 (−310 to −262)	RR 0.055 (0.018 to 0.17)	Low
Change in serum phosphate (mmol/L), primary window	2 (231)				MD 0.36 (0.24 to 0.49)	Low
Serious or severe hypersensitivity, primary window	3 (3147)	2	1	−1 (−2 to 4)	RR 0.32 (0.033 to 3.02)	Very low
Serious or severe hypersensitivity, extended window	2 (377)				Not pooled: Yanamandra 2026 0/140 vs 0/140; PHOSPHARE-IBD 0/48 vs 1/49	Very low
Any hypersensitivity or infusion reaction, primary window	2 (3105)				Not pooled: Bombač Tavčar 2024 1/100 vs 4/100; RAPIDIRON 340/1454 vs 70/1451	Very low
Serious cardiac and vascular adverse events, primary window	2 (242)				Not pooled: PHOSPHARE-IDA04 0/63 vs 0/60; PHOSPHARE-IDA05 0/62 vs 0/57	Very low
Serious cardiac and vascular adverse events, extended window	2 (377)	5		0 (−14 to 15)	RD 0.00011 (−0.014 to 0.015)	Very low
Osteomalacia and fractures	0				No head-to-head RCT reported this outcome	—
Adjudicated cardiovascular events	0				No head-to-head RCT reported this outcome	—

4 Discussion

4.1 Main findings

In head-to-head randomised trials, FDI may reduce the incidence of hypophosphataemia, of severe hypophosphataemia and of hypophosphataemia persisting at the end of the primary window compared with FCM. The effect on the primary outcome was large and consistent across individual trials that used identical doses of the two drugs and trials that used different regimens, although the pooled same-schedule subgroup was imprecise. Its certainty is low because of risk-of-bias concerns in all contributing trials and because the number of events was below the optimal information size, which we applied strictly; the confidence interval excludes the null by a wide margin. All trials contributing to the pooled estimates for hypophosphataemia, persistent hypophosphataemia and change in phosphate were funded by the manufacturer of FDI; we did not downgrade for this, but no independent trial reported the outcome as defined: the two independent trials that measured phosphate used a single late measurement or did not state the threshold. Evidence on hypersensitivity and cardiovascular safety is very uncertain. No head-to-head trial has reported osteomalacia or fractures, the clinical consequences that motivate concern about hypophosphataemia.

4.2 Comparison with previous syntheses

Previous syntheses reached their estimates mainly through network, indirect or study-level comparisons [2–7]. Our review restricts the evidence to direct randomisation. This reduces the number of trials, but it removes the exchangeability assumption between trials against different comparators. The direction of our findings on hypophosphataemia is consistent with the syntheses that addressed it [2, 5, 6]. We do not compare magnitudes, because the populations, thresholds and methods differ.

4.3 Strengths and limitations

Strengths.

- Independent duplicate screening, extraction, risk-of-bias assessment and GRADE rating, with author adjudication of discordances.
- Author verification against the source of every analysed data row.
- Arm-level data taken from registries as well as articles.
- Aggregate numbers produced only by versioned scripts, so every estimate can be traced to a data row and page.

Limitations.

- The protocol was frozen after the repository already held unstructured notes and extractions on some trials, and it was not registered prospectively.
- Several deviations were decided after seeing the results: the main model with zero cells, the correction of the pooling rule and the new risk-difference rule, the application of the threshold rule to severe hypophosphataemia, the inclusion of a double-zero trial in the extended window of the primary outcome, not pooling any hypersensitivity, the descriptive registered-versus-published comparison, and the subgroup classification. All are declared; for zero cells the alternative models are reported, and any hypersensitivity is reported trial by trial.
- Embase, Cochrane CENTRAL and WHO ICTRP were not searched, so trials indexed only there, and registered trials outside ClinicalTrials.gov and the EU registers, may have been missed; the

publication-bias rating is provisional for this reason.

- Few trials contributed to each outcome, and the trials informing the pooled estimates for hypophosphataemia, persistent hypophosphataemia and change in phosphate were all funded by one manufacturer.
- Thresholds for severe and wide-threshold hypophosphataemia differed between trials, and the risk difference for severe hypophosphataemia includes a trial that measured phosphate only at 6 weeks.
- Eligibility evidence sheets were prepared by a single agent run, and the eligibility proposals were approved in blocks.
- The hypersensitivity outcomes rest on heterogeneous definitions.
- Automated agents performed the first pass of every step, and concordant exclusions at title and abstract, as well as concordant risk-of-bias and GRADE judgements, were not reviewed by an author. We mitigated this with duplicate independent runs, algorithmic checks and author adjudication of discordances, but it remains a methodological novelty.

4.4 Implications for research

Future head-to-head trials should:

- be independent of manufacturers;
- define hypophosphataemia at prespecified thresholds and time points;
- follow participants long enough to capture musculoskeletal consequences;
- use standard criteria for serious hypersensitivity.

5 Conclusions

In head-to-head randomised trials in adults with iron deficiency, FDI may reduce the incidence of hypophosphataemia compared with FCM (low-certainty evidence). The comparative effects on hypersensitivity and cardiovascular events are very uncertain, and no head-to-head trial has reported osteomalacia or fractures.

Declarations

Funding. This work received no specific funding.

Competing interests. The authors declare no competing interests.

Ethics. This review used only published or registry-posted aggregate data; ethical approval and consent were not required.

Data and code availability. The frozen protocol, search logs, screening decisions, arm-level extraction files, risk-of-bias and GRADE files, the analysis script (`50-Metanalisi/analisi.R`) and all its outputs, the agent instructions and an export of the selection database are openly available at <https://github.com/filippobuscemi11463-sketch/FDI-FCM-metaanalysis>. Full texts and abstracts of the included reports are not redistributed.

Use of AI tools. Large-language-model agents (Claude, Anthropic, via Claude Code) were used for duplicate screening, extraction, risk-of-bias and GRADE assessments, and for drafting code and text, under the procedures described in Methods. Concordant exclusions at title and abstract screening

and concordant risk-of-bias and GRADE judgements were accepted without author review; eligibility proposals prepared by a single agent run were approved by an author; discordant and uncertain records, extracted values and judgements were decided by an author, who also verified every data row selected for analysis. The authors take full responsibility for the content.

Author contributions. F.B. and P.B. contributed equally to this work. Both authors approved the final manuscript.

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